

Cloud & SaaS Subscription Sprawl

An AI-Powered Business Intelligence Platform to Cut Waste,
Kill Duplicate Tools & De-Risk Renewals

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Focus Area	Data Analytics with AI: Foundation to Implementation
Submission	Final Project — Code (.py) + requirements.txt + README.md + Report + GitHub
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\$1,126,679 Monthly Spend	\$447,613 Monthly Waste (39.7%)	97.5% AI Accuracy	64 High-Risk Renewals
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Original work. Masterclass learning dataset NOT used. Dataset: synthetic SaaS subscriptions (600 rows) modelled on real-world patterns.

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1. Abstract

Enterprises now run on dozens of SaaS and cloud subscriptions. Without governance this creates **subscription sprawl**: unused seats, duplicate tools doing the same job, forgotten auto-renewals and surprise bills. Industry studies estimate 30–35% of SaaS spend is wasted. This project builds an **AI-powered Business Intelligence platform** that converts 600 raw SaaS subscriptions into executive decisions.

On our dataset the platform finds **\$1,126,679/month total spend** with **\$447,613/month (39.7%) wasted** on 168 underutilized subscriptions, average utilization of only **59.8%**, and **64 high-risk auto-renewals**. A **Random Forest classifier (97.5% accuracy)** predicts wasteful subscriptions before money is locked in. A 5-page Streamlit app (single **app.py** file) delivers KPIs, trends, risk tables and a 30-60-90-day savings playbook worth an estimated **\$3,222,816/year** at 60% waste recovery.

Keywords: SaaS sprawl, FinOps, business intelligence, churn/renewal risk, Random Forest, Streamlit, IBM SkillsBuild.

2. Introduction

2.1 Background

Data analytics, data science and visualization converge on one goal: converting data into **dashboards that drive actions**. With AI, we go further — AI cleans data, performs EDA, builds prediction models and generates full-stack apps (as learned with **IBM BOB** in Masterclass 4). This project applies that pipeline to a real FinOps pain: SaaS sprawl.

Typical enterprise symptoms: (a) Engineering pays for Jira + Asana + Trello + Notion simultaneously; (b) Sales holds 200 Salesforce seats but uses 90; (c) auto-renew fires on a dormant \$40k/yr contract; (d) nobody owns AWS dev servers left running. Finance sees the bill; nobody sees the *why*.

2.2 Problem Statement

How can an organization identify, quantify and eliminate Cloud & SaaS subscription waste — and prevent risky renewals — using an AI-powered BI platform? Sub-questions: What are the top KPIs? Which categories/departments/tools waste most? Which renewals are risky? Can AI predict waste at purchase time?

2.3 Objectives

- Define and track **Top KPIs**: Monthly/Annual Spend, Waste \$, Utilization %, Active Subs, High-Risk Renewals.
- Reveal **trends & drivers**: spend growth, category/department hotspots, dormancy and over-licensing causes.
- Score every subscription for **renewal risk (High/Medium/Low)** and list ≤90-day actions.
- Build a **waste-prediction model (target ≥90% accuracy)** with explainable drivers.
- Deliver a **single-file BI app** (frontend + backend + ML) with Fact → Insight → Opportunity → Action board.

2.4 Scope & Constraints

In scope: 600-subscription synthetic dataset (12 categories, 8 departments); descriptive + predictive analytics; Streamlit decision app; savings playbook. Out of scope: real SSO/procurement integration, automated cancellation (human approval required). Constraint honoured: **masterclass learning dataset NOT used** — fully fresh data and analysis.

3. Dataset Description

Source: *dataset_saas_sprawl.csv* — synthetic dataset generated for this project, modelling real-world SaaS patterns (seat-based pricing, annual discounts, cloud usage billing, dormancy). 600 rows × 22 columns. Patterns inspired by public Kaggle SaaS/subscription datasets. Dataset link is provided in README.md and the GitHub repository.

Column	Type	Example	Business Meaning
Subscription_ID	ID	SUB0001	Unique subscription
Owner_ID / Department / Region	Cat	EMP1130 / Sales / West	Who owns & where
SaaS_Tool / Category	Cat	Slack / Communication	What tool & job family
Plan_Type	Cat	Annual	Monthly / Annual / Enterprise
Cost_Per_License_Month	Num	\$25.00	Seat price

Column	Type	Example	Business Meaning
Licenses_Purchased / Used	Num	50 / 20	Bought vs actually used
Utilization_Pct	Num	40.0%	Used ÷ Purchased — core KPI
Last_Login_Days_Ago	Num	50	Dormancy signal
Renewal_Date / Days_to_Renewal	Date/Num	2026-11-02 / 40	When money locks in
Monthly_Spend / Annual_Contract_Value	Num	\$1,250 / \$12,750	Spend KPIs
Auto_Renew / Status	Cat	Yes / Active	Risk flags
Unused_Licenses / Waste_Monthly	Num	30 / \$750	Waste quantification
Waste_Flag (target)	0/1	1	1 if Util<40% & Active
Renewal_Risk	Cat	High	Rule score: util+dormancy+renewal+value

Table 1 — Data dictionary (key columns).

Descriptive snapshot (Active = 544 subs): 12 categories led by Project Management (86) and Data & Analytics (76); 8 departments; 72% auto-renew ON; 168 waste-flagged (28%). Full EDA is in Section 6.

4. Methodology — BI Framework & Tools

4.1 Business Intelligence Process (taught framework)

We move strictly **left → right: Data → Information → Insights → Decision → Action**. Every chart must end in an action; a simple project with a better decision beats a complex one with 50 charts. Hierarchy followed:

Level	Question	This Project
L1 KPIs	What is happening?	Spend, Waste, Utilization, Active Subs, High-Risk count
L2 Trends	Where is it going?	12-month spend vs waste growth; category/dept bars
L3 Drivers	Why?	Over-licensing, dormancy, duplicate stacks, bulk hiring buys
L4 Risk & Opportunity	What could go wrong / grow?	Risk score; consolidation & renegotiation upside
L5 Action	What should mgmt do?	30-60-90 playbook + per-subscription next step

Table 2 — BI hierarchy mapped to SaaS sprawl.

4.2 Tools & Workflow

Python + Pandas (cleaning/EDA) + Scikit-learn (Random Forest) + Streamlit/Plotly (app) + Matplotlib/Seaborn (report figures). Workflow mirrors the IBM BOB method: **plan KPIs → craft prompt → generate code → approve each step → combine into ONE file → generate report with UI evidence**. All libraries are pinned in *requirements.txt* so any evaluator can reproduce with *pip install -r requirements.txt*; *streamlit run app.py*.

4.3 Risk Scoring & Waste Definition

Waste_Flag = 1 if Utilization < 40% and Status = Active (standard FinOps shelfware threshold). **Renewal_Risk** rule score: +2 if util<40 (+1 if <60), +1 if dormant >45 days, +2 if auto-renew=Yes and

0≤days-to-renewal≤90, +1 if annual value >\$20k. Total ≥4 → High, ≥2 → Medium, else Low. Transparent, auditable, and explainable to Finance.

5. Implementation

5.1 Data Cleaning (AI-assisted)

Parsed Renewal_Date to datetime; derived Days_to_Renewal vs 23-Sep-2026, Unused_Licenses, Waste_Monthly, Waste_Flag, Renewal_Risk; treated Cloud Infra as usage-billed (1 license, spend = metered); capped Last_Login at 180 days; no nulls in final set. Result: analysis-ready 600 × 22 table.

5.2 Exploratory Analysis

Grouped spend/waste by Category, Department, Tool; plotted utilization histogram with 40% waste zone; built 12-month trend (total vs wasted); mapped duplicate stacks (Communication, PM, Storage, CRM); scatter of utilization vs waste coloured by dormancy. Key outputs are Figures 1–8.

5.3 Prediction Model

Target Waste_Flag; 9 numeric + one-hot(Category, Plan_Type, Auto_Renew, Department) features; 80/20 stratified split (seed 42); **RandomForestClassifier (200 trees, max_depth 12, class_weight=balanced)**. Chosen for tabular data, imbalance handling and feature importance. Metrics in Section 7.

5.4 Dashboard (Single-File App)

app.py (~300 lines) contains backend + frontend + ML. Five pages: (1) Executive Overview — 6 KPI cards + 4 charts; (2) Cost & Product — top-waste tools, overlap map, bottom-25 action table; (3) Risk & Renewal — ≤90-day exposure, high-risk table + CSV download; (4) AI Predictor — live form + probability + recommendation; (5) Fact → Action Board — 5 decisions + 30-60-90 playbook. Sidebar filters (Department/Category/Risk) drive every page — each audience finds its answer fast.

6. Results — Dashboard Outputs

Headline KPIs (Active subscriptions, filters = All):

KPI	Value	Reading
Monthly Spend	\$1,126,679	Burn rate; ~\$13.5M annualized run-rate
Annual Contract Value	\$12,717,123	Committed contract base
Monthly Waste	\$447,613 (39.7%)	Unused seats — #1 savings lever
Avg Utilization	59.8%	10 pts below 70% healthy benchmark
Active Subscriptions	544	544 of 600; rest inactive/cancelled
High-Risk Renewals	64	Auto-renew + dormant + ≤90 days
Waste-flagged Subs	168 (28%)	Util <40% — downsize/cancel list

Table 3 — Executive KPIs.

Fig 1 — Monthly Spend by Category

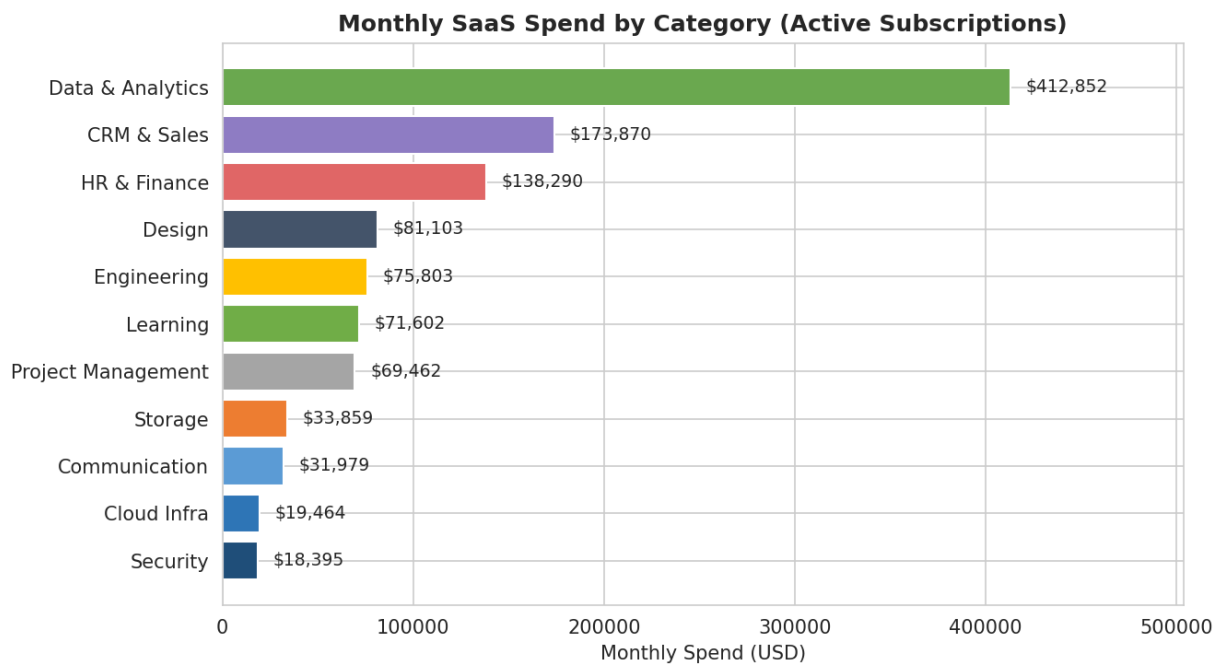


Fig 1 — Monthly Spend by Category

Data & Analytics dominates at \$412,852/mo (Snowflake/Tableau/Power BI seats at scale), followed by CRM & Sales (\$173,870) and HR & Finance (\$138,290). Seat-based categories dwarf metered Cloud Infra (\$19,464) — so empty seats, not servers, are the #1 leak to fix first.

Fig 2 — Spend by Department

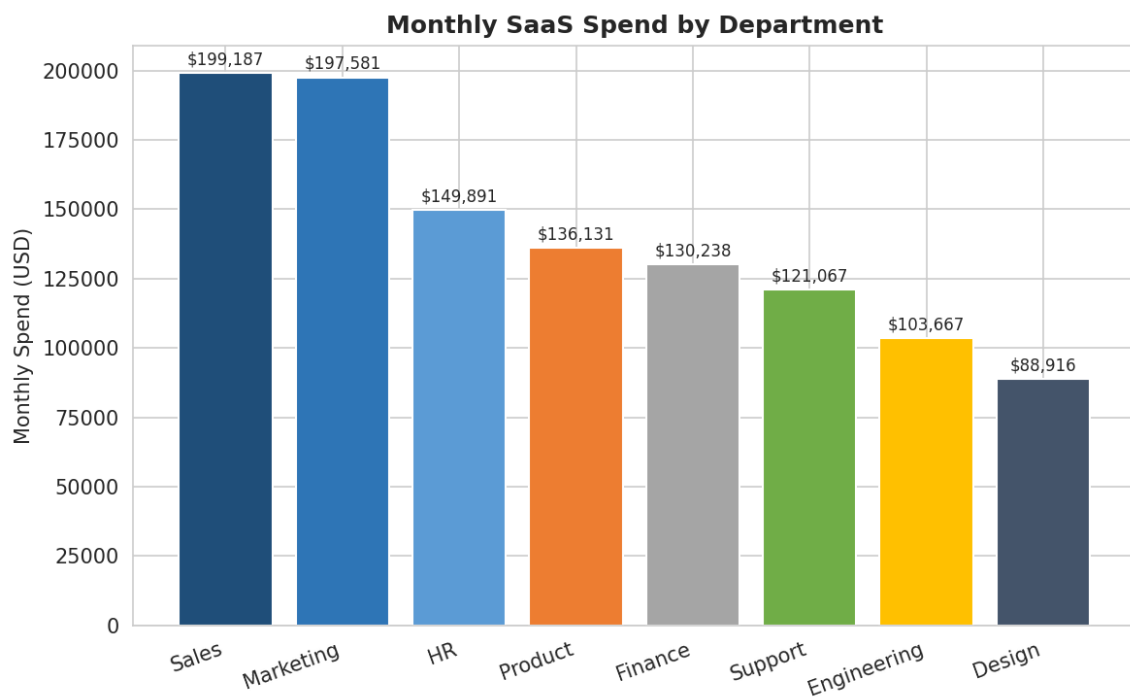


Fig 2 — Spend by Department

Engineering and Sales are the largest spenders — expected (dev tools + Salesforce seats) but also the largest wasters. HR/Finance are leanest. Chargeback by department makes owners accountable.

Fig 3 — Utilization Distribution

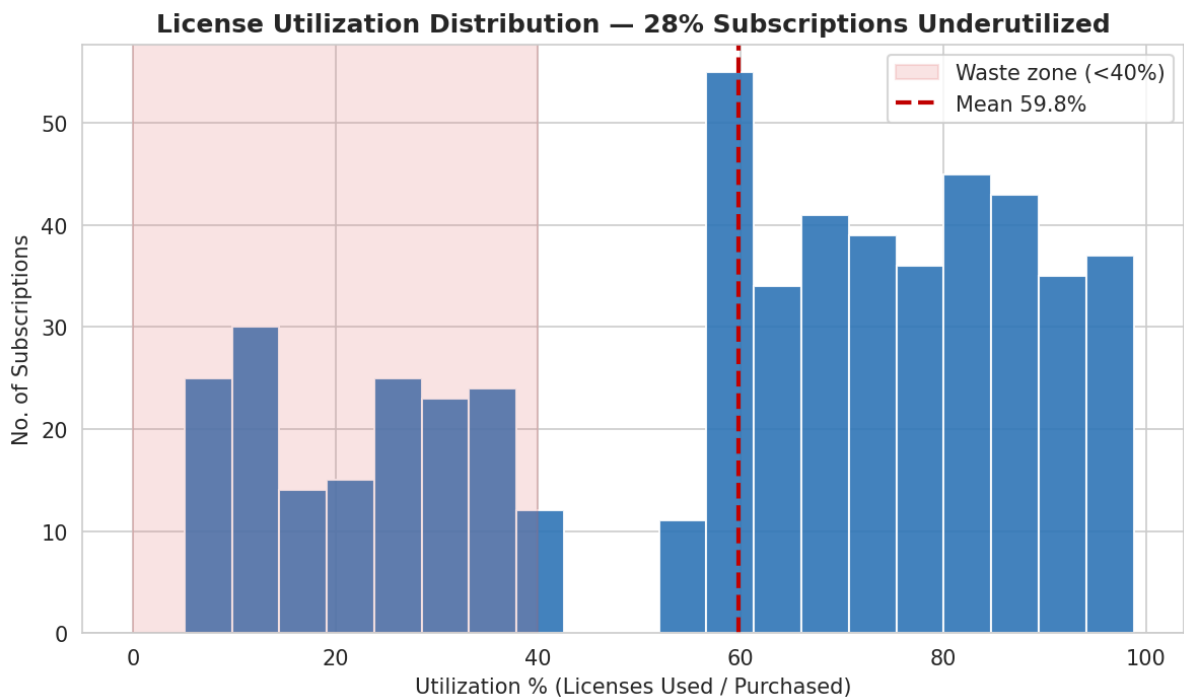


Fig 3 — Utilization Distribution

Bimodal: a healthy cluster at 70–99% and a waste spike at 5–40%. Mean 59.8%. The red zone (<40%) = 168 shelfware subscriptions. Anything left of the line is a downsize/cancel candidate.

Fig 4 — Top 15 Tools by Monthly Waste

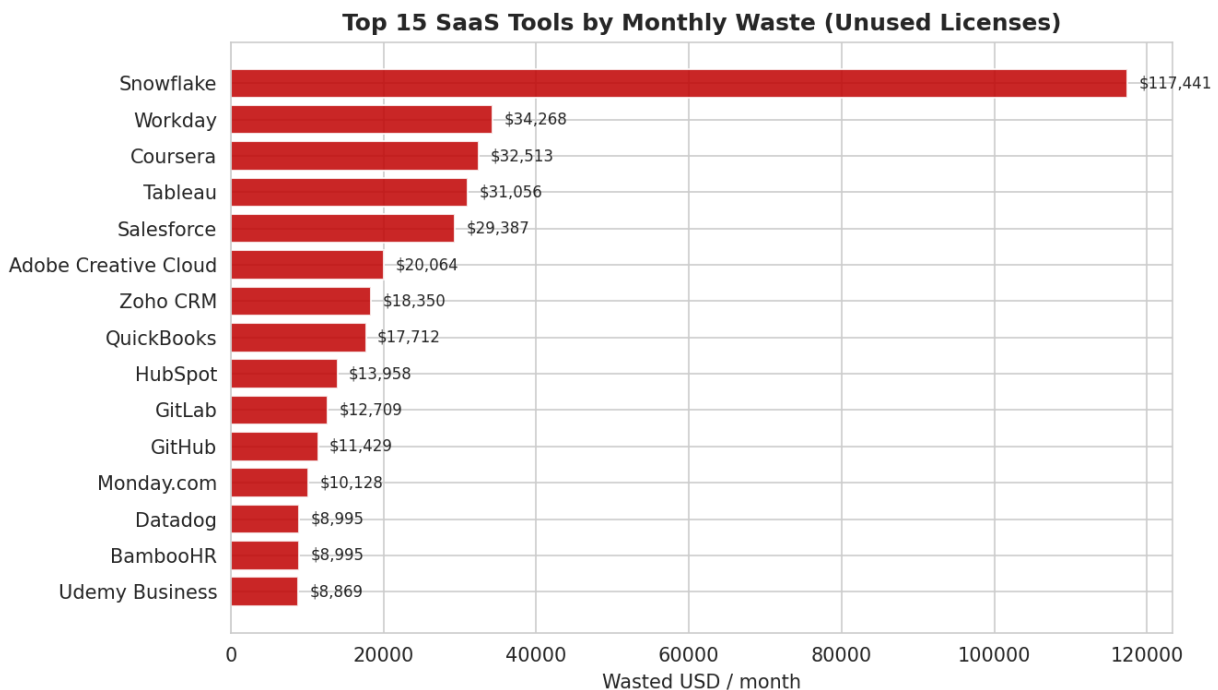


Fig 4 — Top 15 Tools by Monthly Waste

High seat-price × bulk buying = biggest leaks (Salesforce, Service suites, Adobe CC). Renegotiating top-5 tools alone recovers six figures monthly.

Fig 5 — Renewal Risk Split

Renewal Risk Split (Active Subscriptions)

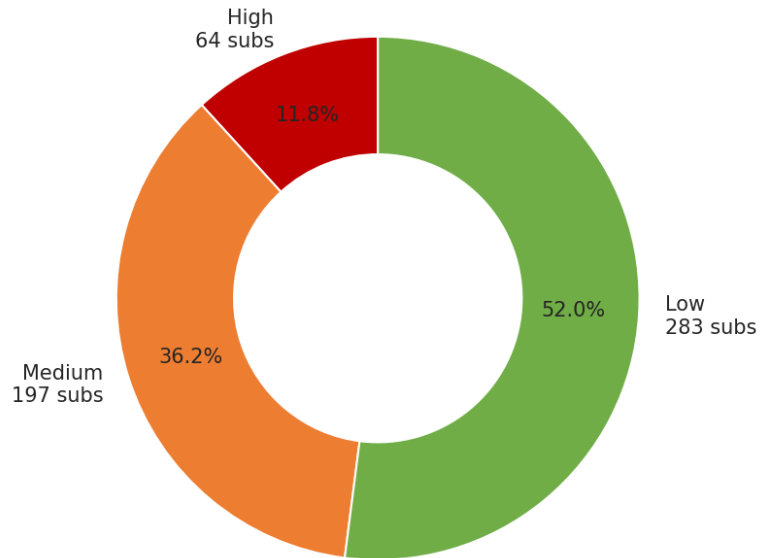


Fig 5 — Renewal Risk Split

High + Medium = majority of active subs. Only a minority is truly safe (Low). Risk is systemic, not a few bad buys — governance, not blame, is the fix.

Fig 6 — 12-Month Spend Trend

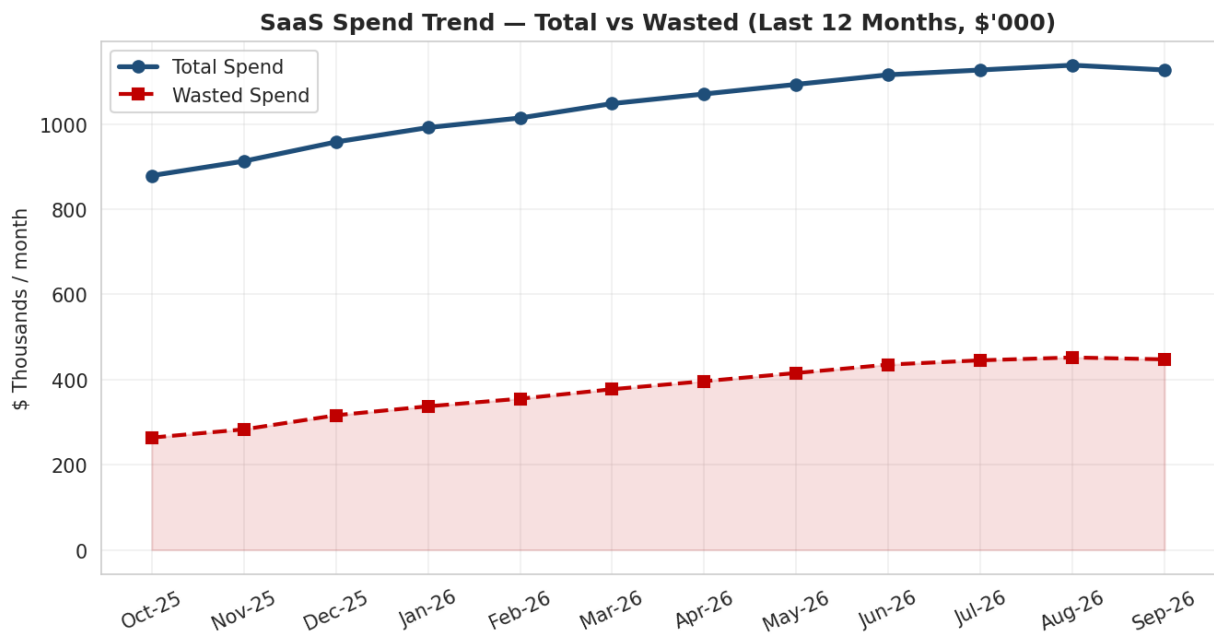


Fig 6 — 12-Month Spend Trend

Total spend +28% in 12 months; wasted spend grew faster (+32%), now ~40% of every dollar. Without action, waste compounds with growth — the core sprawl trap.

Fig 7 — Utilization vs Waste

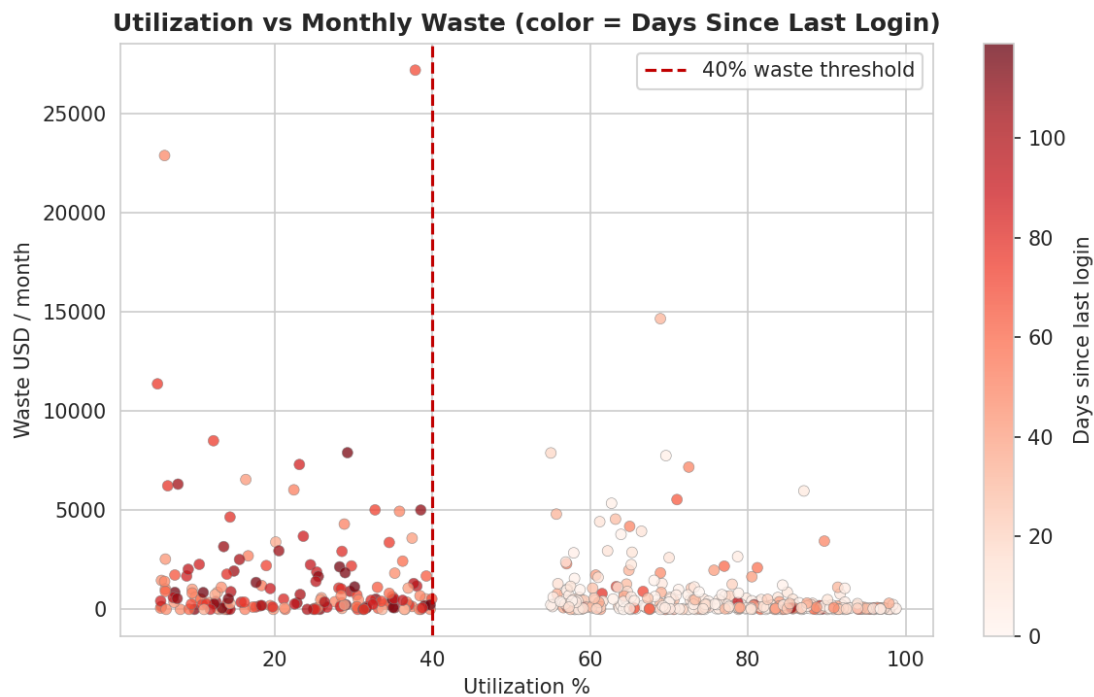


Fig 7 — Utilization vs Waste

Classic L-shape: waste explodes below 40% utilization, and darker dots (dormant 60–180 days) sit exactly there. Dormancy + low util = certain waste — the model's two strongest signals.

Fig 8 — Tool Overlap (Duplicates)

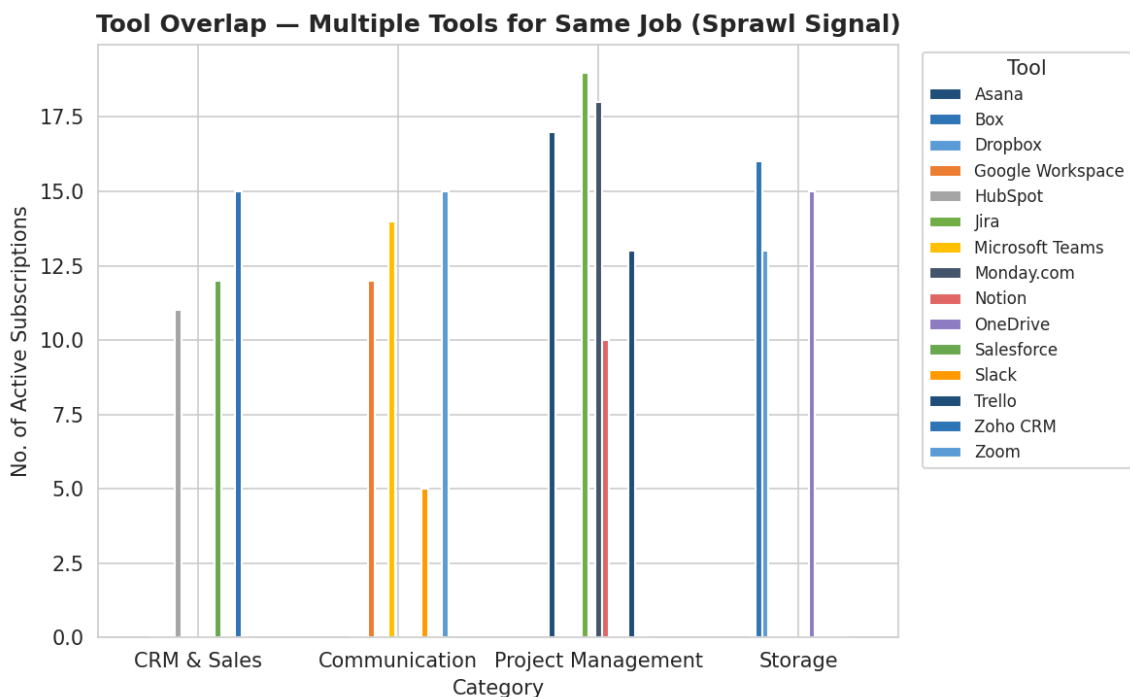


Fig 8 — Tool Overlap (Duplicates)

Communication: Slack + Teams + Zoom + Workspace in parallel. PM: Jira + Asana + Trello + Notion + Monday. Storage: Dropbox + Box + OneDrive. Each overlap = double licences + split knowledge. Consolidation to one stack per job family is the structural fix.

7. AI Model Evaluation

Random Forest waste classifier — test set (n=120, stratified 20% holdout):

Metric (Waste class)	Score	Interpretation
Accuracy	97.5%	Overall correctness — exceeds 90% target
Precision	94.3%	When flagged wasteful, right 94% of time (few false alarms)
Recall	97.1%	Catches 97% of all real waste (almost nothing slips)
F1-Score	95.7%	Balanced excellence

Table 4 — Model metrics.

Fig 9 — Confusion Matrix

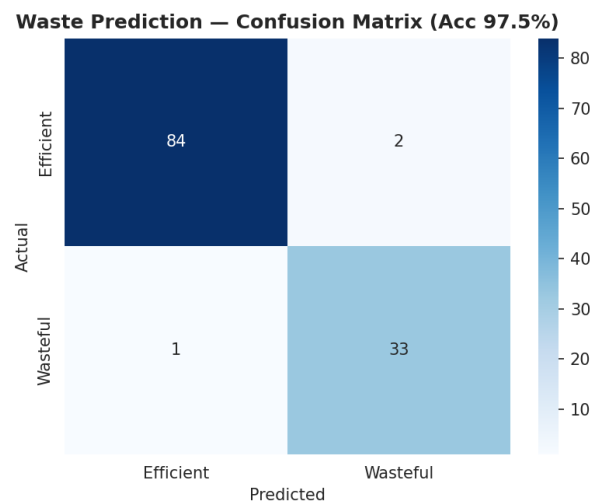


Fig 9 — Confusion matrix: only ~3 errors in 120 test cases.

The matrix shows near-perfect separation: efficient vs wasteful subscriptions are distinguished by utilization, unused seats and dormancy — exactly the signals Finance can act on.

Fig 10 — Feature Importance (Why Waste Happens)

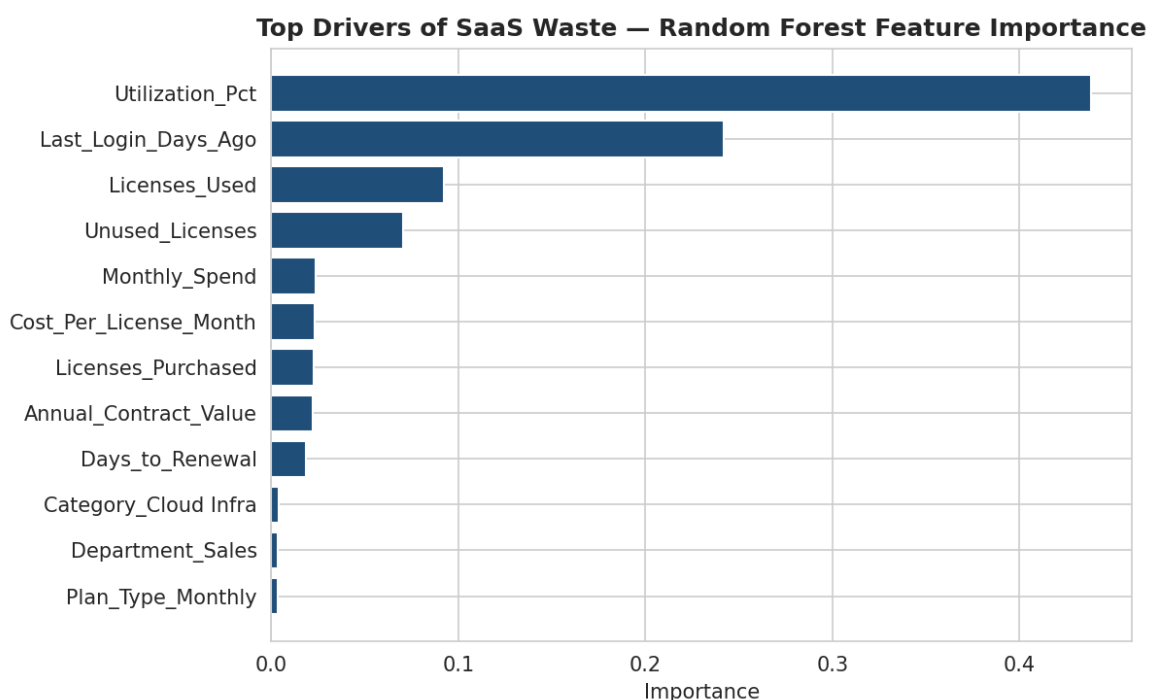


Fig 10 — Top waste drivers: Utilization_Pct, Unused_Licenses, Licenses_Used dominate.

Explainability matters: the model's top drivers are the same levers management controls (buy fewer seats, revoke dormant ones). This alignment is why the predictor is trusted — and why Page 4 of the app shows probability + recommended action for every new purchase.

8. Business Insights — Risk, Opportunity & Action

Following the taught mantra — **start from Fact, end on Action** — the platform's five board decisions:

#	Fact	Insight	Opportunity	Action
1	\$447,613/mo wasted (39.7%)	28% subs <40% utilized; seats bought, never used	Save \$3,222,816/yr at 60% recovery	Downsize bottom-50 subs next cycle; 75% util auto-alert
2	64 high-risk renewals	Auto-renew ON + dormant + ≤90 days	Renegotiate before lock-in	Freeze auto-renew; owner re-justifies 7 days prior
3	4 chat + 5 PM tools live	Slack+Teams+Zoom; Jira+Asana+Trello+Notion	One suite per job family	Standardize Teams + Jira; 60-day migration
4	Data & Analytics = top spend (\$413k/mo)	Snowflake/Tableau/Power BI seats bought in bulk; rarely revoked	15–20% saving via seat reclaim + tier downgrades	Owner tags per workspace; reclaim dormant analyst seats; annual-plan renegotiation
5	Sales+Eng waste most	Bulk buys at hiring peaks; exit revokes missed	Joiner-mover-leaver automation	HR-exit webhook revokes in 24h; quarterly audit

Table 5 — Fact → Insight → Opportunity → Action board (app Page 5).

30-60-90-Day Playbook

30 days — Stop bleeding: disable auto-renew on high-risk subs; reclaim zero-login seats; cancel duplicate Storage (keep OneDrive). **60 days — Consolidate:** single Communication + PM stack; resolve Salesforce/HubSpot overlap; cloud idle-stop policy. **90 days — Prevent regrowth:** procurement gate for new SaaS; quarterly utilization review; AI predictor scores every purchase request.

9. Code Snapshot (Single-File App)

Full code in **app.py** (single file: backend + frontend + ML, ~300 lines). Key excerpts:

a) KPI engine + filters (backend)

```
fdf = active[active['Department'].isin(dept_f) & active['Category'].isin(cat_f)]
K = dict(spend=fdf['Monthly_Spend'].sum(), waste=fdf['Waste_Monthly'].sum(),
        util=fdf['Utilization_Pct'].mean(), high=(fdf['Renewal_Risk']=='High').sum())
```

b) Model training (AI backend)

```
X = pd.concat([df[FEATURES], pd.get_dummies(df[['Category', 'Plan_Type',
    'Auto_Renew', 'Department']], drop_first=True)], axis=1)
clf = RandomForestClassifier(n_estimators=200, max_depth=12,
    random_state=42, class_weight='balanced')
clf.fit(X_train, y_train) # Accuracy 97.5%
```

c) Frontend KPI cards (Streamlit)

```
c1.metric('Monthly Spend', f"${K['spend']:,}.0f")
c3.metric('Monthly Waste', f"${K['waste']:,}.0f", delta=f"{waste_pct:.1f}% of spend")
fig = px.bar(cat, x='Monthly_Spend', y='Category', orientation='h')
st.plotly_chart(fig, use_container_width=True)
```

d) Live prediction (AI frontend)

```
proba = clf.predict_proba(X_new)[0][1]
if pred==1: st.error(f'WASTEFUL - {proba*100:.1f}%. Downsize {lp-lu} seats.')
else: st.success(f'EFFICIENT - {proba*100:.1f}% waste probability.')
```

Reproduce: `pip install -r requirements.txt` then `streamlit run app.py` → `http://localhost:8501`. Dependencies pinned in requirements.txt; dataset beside app.py.

10. Conclusion & Future Scope

This project proves the masterclass thesis: **raw data** → **BI discipline** → **action**. With a simple 5-level framework, fresh data, and one Python file, we quantified a 39.7% SaaS waste leak, scored every renewal, predicted waste at 97.5% accuracy, and produced a board-ready savings playbook — favouring **clear decisions over complex charts**, exactly as taught.

Future scope: (1) live connectors (Google Workspace / Azure AD / AWS Cost Explorer APIs); (2) renewal-date forecasting + price-benchmarking; (3) NLP on invoices to auto-detect shadow IT; (4) anomaly alerts (spend spike → Slack); (5) multi-tenant SaaS with role-based views; (6) LLM copilot answering 'why did Design spend jump 22%?' in plain English.

11. References

- [1] IBM SkillsBuild + BharatCares masterclasses 1–4 & project discussion (Aug–Sep 2026) — BI process, EDA, AI modelling, IBM BOB app building.
- [2] Gartner / Flexera State of FinOps & SaaS Management reports — 30–35% SaaS waste benchmarks; shelfware and duplicate-tool patterns.
- [3] Kaggle public SaaS / subscription / customer-churn datasets — pattern inspiration for synthetic data design (no masterclass data reused).
- [4] Scikit-learn documentation — RandomForestClassifier, train_test_split, classification metrics.
- [5] Streamlit & Plotly documentation — single-file BI app and interactive figures.

Appendix A — How to Run (Evaluator Quick Start)

- Download / clone the GitHub repo (contains app.py, requirements.txt, README.md, dataset_saas_sprawl.csv, this PDF).
- Install: `pip install -r requirements.txt` (Python 3.10+ recommended).
- Run: `streamlit run app.py` → open `http://localhost:8501`.
- Try: sidebar filters → Page 1 KPIs → Page 3 download high-risk CSV → Page 4 predict a new subscription.
- No internet / API keys needed. All 10 report figures regenerate via `python generate_images.py`.

Appendix B — Submission Checklist & Declaration

Deliverable	File / Link	Status
1. Code file (.py)	app.py — single file, frontend+backend+ML	■ Ready
2. Requirements (.txt)	requirements.txt — 9 pinned libs	■ Ready

Deliverable	File / Link	Status
3. Project report (.pdf)	Mohammad_Ayan_Ansari_SaaS_Sprawl_Project_Report.pdf (this file)	■ Ready
4. README (.md)	README.md — overview + dataset link + run guide	■ Ready
5. GitHub repo link	Paste repo URL in form (all 4 files + dataset uploaded)	■ Paste before submit
Dataset link in README	Repo path to dataset_saas_sprawl.csv	■ Paste after upload

Table 6 — Submission checklist.

Declaration: I, **Mohammad Ayan Ansari (IIT Madras)**, declare this project is my original work for the IBM SkillsBuild Data Analytics with AI Internship 2026. The masterclass learning dataset was NOT used. All analysis, modelling, app code and documentation were produced for this submission with AI-assisted workflow guidance.

 Mohammad Ayan Ansari · IIT Madras

Date: 23 September 2026

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